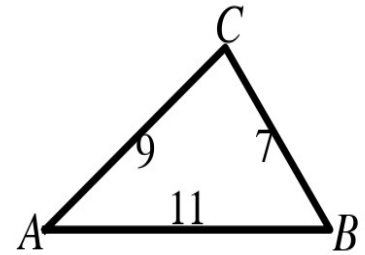


Name: _____

Use the figure at right for 1. Then order as directed.

1. Order the angles of $\triangle ABC$ from smallest to largest.
2. In $\triangle DEF$, $m\angle D = 48^\circ$, $m\angle E = 79^\circ$, and $m\angle F = 53^\circ$. Order the sides from shortest to longest.



Can the three lengths be the sides of a triangle? Show the check.

3. 5, 9, 13

4. 4, 6, 11

5. 8, 8, 16

Find the range of possible lengths for the third side.

6. Two sides measure 7 and 18.

7. Two sides measure 12 and 12.

Indirect proof.

8. Write the first sentence of an indirect proof of: "A triangle has at most one obtuse angle."
9. Finish that proof.
10. In $\triangle PQR$, $m\angle P = 40^\circ$ and $m\angle Q = 75^\circ$. Which side is longest? Explain.